

- Hello, welcome back. As you know, my name is Nicole Meier. I'm happy to be back here with you. We're gonna talk about the last of the DIGIN dysfunctions, nervous system, and in this case, we're gonna talk about the enteric nervous system, manifestations, mediators, and mitigators. I have no disclosures. This is a list of your learning objectives that you can review. And this is a list of the IFM clinical skills and performance objectives from today's segment. So, we are on nervous system, as I said, or more specifically for the purposes of this talk, enteric nervous system, that part of the gut-brain axis that you are gonna understand by the end of this segment. Our focus in this section is reviewing the physiology and features of the gut-brain axis and the enteric nervous system, as well, we're going to talk about and reach an understanding of the enteroendocrine system and its clinical manifestations. So to orient you in time and space, we are talking about the central nervous system and the peripheral nervous system. The peripheral nervous system, of course, has an autonomic nervous system component. That autonomic nervous system component was always taught to me many years ago as having a sympathetic and parasympathetic branch only. But what we know is that there's an enteric nervous system branch. It's a third component of the autonomic nervous system. Here are some definitions that we're going to go over about the gastrointestinal components of the nervous system. So first, the gut-brain axis. This is the coordinated response between the enteric nervous system, the autonomic nervous system, and the HPA axis via some of these neuro-immune-endocrine mediators that we're going to speak of. The enteric nervous system is part of the larger gut-brain axis. What we're talking about exactly when we say gut-brain axis, I like to think of it as wires between telephone poles, I know I'm dating myself here, or a large freeway with many lanes, and in these lanes, there are many vehicles. So these lanes might be the CNS, the central nervous system, the autonomic nervous system, the enteric nervous system, the microbiota, et cetera. And the messages, the cars zipping back and forth, can be immune, they can be neural, they can be hormonal. The enteric nervous system is the axonal network providing a direct connection between the CNS and the entire length of the GI tract. It modulates central nervous system functions and it modulates various GI functions. The vagus nerve is another big player in the gut-brain axis, and that is the 10th cranial nerve, as you know. It's a specific nerve that acts as a major conduit within this axis, transmitting signals between the brain and the gut. The enteric nervous system is sometimes referred to as the second brain, the axonal nervous system between the central nervous system and the gut. But it does function independently of the central nervous system. It is the translator that takes a message from the microbiota and communicates it to the central nervous system and the GI tract. The gut-brain axis is the highway, as we mentioned, the lanes are all the different systems involved, and the vehicles are the messages. So this image demonstrates the bidirectional nature of the gut microbiota and the CNS, central nervous system. I think a great example of the CNS affecting microbiota is how chronic stress or even a brain injury can

impact the microbiome. So there are changes in the microbiota as a result of these things happening. Another good example of the microbiota affecting the central nervous system is diversity changes related to neurodegenerative diseases like Parkinson's disease, or mental health disorders like depression and anxiety because there is microbiome alteration in neurotransmitter change. The enteric nervous system has normal physiological functions that we're going to review. Most importantly, please remember that the enteric nervous system functions independently of the central nervous system because of something called the autonomous myenteric plexi. The NS controls motility via the migratory or migrating motor complex. It mixes contents and it propulsive material forward slowly or not so slowly backward as is the case in vomiting when you're trying to get rid of noxious substances. It modifies nutrient handling by modulating blood flow and barrier function. It manages secretions of things like gastrin, digestive enzymes, and bioactive peptides. The enteric nervous system regulates gut immunity by sensing and responding to the intestinal environment, influencing immune cell activity and barrier function through neurotransmitters and neuropeptides. The vagus nerve provides a direct connection between the GI tract and the central nervous system. As part of the parasympathetic system, the vagus nerve innervates many organs, and it communicates between the brain and the ENS. The vagus nerve innervates the gut lining, but it does not come into direct contact with the microbiota. It's separate from the enteric nervous system, but critically important for communication. The enteric nervous system extends from the lower esophagus to the rectum. It's a web of sensory, motor, and interneurons that are in the bowel wall. It has a submucosal layer that's responsible for secretion and absorption and a myenteric layer that houses the motor complex that we spoke about. While the communication between the central nervous system and the enteric nervous system is bidirectional, the enteric nervous system can function independently, particularly when you talk about peristalsis or the motor functions, blood flow, and secretions. These are independent functions of the enteric nervous system. The take home message here on this slide is that there is bidirectional crosstalk in the gut-brain axis. There's crosstalk between the CNS, the ANS, the hypothalamic pituitary axis, the microbiota. Dysbiosis via the enteroendocrine system, which we're gonna talk a little more about later, can influence this HPA axis and the other way around. Stress hormones like cortisol influence gut permeability and mobility, leading directly to symptoms of IBS. Neuropsychiatric disorders such as stress, anxiety, and depression are connected to microbial neurotransmitter production. And HPA axis dysregulation from stress, from trauma can modulate immunity and cause inflammation, which leads to further disruptive barrier function and more immune activation. Revisiting this connection between the gut microbiota and mental health, we see here on this slide the essential production of short chain fatty acids in serotonin. Short chain fatty acids stimulate all sorts of cell types in the gut lining, including those EEC cells or enterochromaffin cells that we spoke about earlier,

which promote the enzyme tryptophan hydroxylase to convert tryptophan to 5HDP to serotonin, and 90% of the body's serotonin is produced this way. Given that melatonin is produced from serotonin, you can see why we emphasize the importance of short chain fatty acids. Gut serotonin doesn't just signal the central nervous system, it modulates motility and neuronal development and differentiation. There are afferent and efferent fibers as we've spoken about already in the gut. There are more afferent than efferent, meaning that the microbiota send more messages to the brain than the brain sends to the gut, which is fascinating really. It includes messages of metabolism, it includes messages of satiety, of things like inflammation. Efferent fibers mostly control motor function. So from the brain to the gut, primarily, it's motor control. While afferent fibers tell us a lot more about what's going on in the gut, they tell us about hunger, they tell us about satiety. There is a bidirectional nature of gut-brain interactions. Afferent chemical signals from the gut travel to the brain via the enteric nervous system by the vagus nerve and then onto the brain. Chemical messengers include dopamine, 50% of which is produced in the gut, serotonin, 90% of which is produced in the gut, norepinephrine, GABA, and more. Clearly, if we wanna influence neurology in our patients, we have to consider this. Not all things, good things follow this pathway, the afferent pathway, sometimes we can have misfolded proteins such as alpha synuclein protein that travel up to the brain in Parkinson's disease for instance. Efferent signaling via the vagus nerve innervates all layers of the gut lining. These nerve fibers are not in direct contact with gut microbiota, but the microbiota do influence the modulation and development of the enteric nervous system neurons. So there is a lot of communication going on in the gut-brain axis. There's a lot of crosstalk, lots of opportunity for messages to change depending on many, many factors. Now the gut also acts as an endocrine organ due to these enteroendocrine cells that secrete gut peptides. They can influence digestion, they can influence appetite and metabolism. So what are these peptides that we're talking about? CCK or cholecystokinin, 5-hydroxytryptamine, glucagon-like peptide-1, ghrelin, and leptin. These are the important peptides that are part of this enteroendocrine system. So this graphic shows us all of the different roles of these peptides, and I'm gonna go through them with you. So CCK or cholecystokinin is excreted from these enteroendocrine cells in the duodenum, in the jejunum in response to eating fat and protein. But also it's secreted when it is triggered by pH changes, including those caused by hydrochloric acid in the stomach. CCK slows gastric emptying. It stimulates excretion of pancreatic proteases, leptin, which is your satiety hormone, and bile acids, and we talked about how important those bile acids are for absorption of your fat-soluble vitamins. CCK has a fascinating effect on your neurology. It increases BDNF, interestingly, brain-derived neurotrophic factor, and therefore memory. Short chain fatty acids increase CCK and therefore satiety. Lipopolysaccharides or LPS as represented in this diagram, which are the active fragment of gram-negative bacteria, they increase CCK,

which then modulates inflammation via NF Kappa B. CCK appears to modulate amyloid plaque development even. Ghrelin or the hunger hormone is primarily produced in the stomach, but also in the small intestine. It communicates with the nervous system crossing the blood-brain barrier, slow energy expenditure, and regulate food uptake. As we have mentioned multiple times, serotonin produced by the gut in the enteroendocrine cells is substantial, 90%. The amino acid tryptophan, precursor to serotonin, is of notable importance here too. The systemic manifestations of tryptophan and serotonin are displayed on this slide. They're important. Anxiety, depression, addiction, influences and sexuality, thermal regulation, diabetes even. What about this very popular peptide that we all know, GLP-1? GLP-1 is produced in the small intestine by these enteroendocrine L cells in response to fat especially or glucose entering that space. It lowers blood glucose and it stimulates insulin secretion in the right conditions, and you can see those right conditions on the right hand side of the slide. When stress goes up, sympathetic activity is elevated on the left hand side of the slide, GLP-1 is inhibited. The vagus nerve obviously plays a big role in regulating GLP-1 and sensing via vagal afferent neurons thereby directly affecting satiety, glycemic control, and energy homeostasis. So we're gonna shift gears a little bit with the clinical focus now. What are the clinical findings when there are gut-brain axis problems, how do you recognize it? What does it look like in your office? So this is just some examples of what we might see in a clinical setting that would cause us to suspect that we have a gut-brain axis problem or dysfunction. We could see difficulty swallowing or dysphasia. Achalasia specifically is a motility disorder of the lower esophageal sphincter that can be due to a loss of inhibitory neurons within the enteric nervous system, resulting in a failure of the lower esophageal sphincter to relax. We might see stool changes like slow transit constipation. Slow transit constipation is a disruption of the ENS, the autonomic nervous system, and the neuroendocrine system, resulting in decreased innervation of the myenteric plexus, leading to dysmotility of the colon that is unresponsive to dietary change or to laxatives. You've all seen these patients in your office where you really struggle to get their bowels moving. Weight gain as a result of chronic stress and dysregulation of these peptides that inform appetite, GLP-1 CCK, ghrelin, leptin, any of these conditions might warrant stool testing if things aren't getting better. In a patient with gut-brain axis dysfunction, we might also just see irritable bowel syndrome, depression and anxiety, autoimmunity. You're getting the point that this is why we start with the gut. Let's talk about something that we see in our offices every day, functional dyspepsia. So how does functional dyspepsia start, and how does it connect to metabolic syndrome? This is a complicated slide, but it is all here for your reference. The impact of chronic stress on the ANS, autonomic nervous system, and the HPA axis can trigger the enteric nervous system, resulting in a change in eating behavior, appetite changes, cravings. There can be changes in gut motility, visceral hyperactivity, and increased inflammation driving

this cluster of symptoms that we call functional dyspepsia. At the same time, this dysfunction in the hypothalamic pituitary axis and the autonomic nervous system promote metabolic syndrome by changing the gut microbiome and driving dietary choices and preferences and promoting inflammation. Okay, so here we have enteric nervous dysfunction as a mediator, and we're gonna set this up on a matrix for you so you understand what it looks like. So I'm gonna populate this matrix first for ease. So the enteric nervous system acts as a mediator to many chronic health concerns. The matrix that we populate when the ENS is dysfunctional looks like this. Under the assimilation node, we might see constipation. This is this dysfunctional migratory motor complex. We might see bloating, heartburn, might involve the vagus nerve not managing lower esophageal sphincter function and gastric secretions. Heavy caffeine use by this patient might increase sympathetic tone, which downregulates vagal function and irritates the gastric mucosa mechanically and impairs parietal cell function. Maldigestion issues due to decreased vagal tone and hypochlorhydria likely contribute to this patient's ankle edema due to protein maldigestion. And poor exercise recovery could be a consequence of something like anemia from a vegetarian diet. We might see early signs of hormone dysregulation due to the development of metabolic syndrome driven by this dysbiotic gut, activation of stress hormones in the HPA axis affecting the enteric nervous system resulting in poor food choices. These alterations in the hormones of appetite and satiety can cause altered metabolic products that drive appetite and food choices in the wrong direction, not to mention inflammation. We might see the development eventually of autoimmune disease under the defensive repair node or chronic inflammatory conditions due to the upregulation of the gut system and the inflammation that is created by this dysbiotic environment communicating via the enteric nervous system to the central nervous system. Allergies and autoimmunity are partially a consequence of gut dysbiosis and immune upregulation. Inflammation breaks down these tight junctions that Swathi was talking about, and the result is immune dysregulation. So now we're going to focus on stress as an antecedent trigger and mediator of gut-brain axis dysfunction via the enteric nervous system. Dysbiotic change is really one of the major modes of action or mechanism of actions whereby the gut-brain axis is altered. The enteric nervous system is exquisitely sensitive to modifiable lifestyle factors. So this is where functional medicine can really improve health by addressing these areas, sleep, exercise, nutrition, stress, relationships. We know that sleep disturbances can influence and be influenced by the enteric nervous system via reduced serotonin melatonin production and unregulated ghrelin and blood sugar disturbances. We know that overexercising or underexercising can increase or activate the HPA axis, which communicates directly with the entire nervous system. And ultra-processed diet low in nutrients, high in inflammatory foods, shapes the microbiome and drives up inflammatory mediators and alters satiety hormones which send signals to the brain to overeat. Stress modulates the enteric nervous system by impacting motility, microbiota, and

therefore barrier function and hormone production. Loneliness, lack of support is associated with, interestingly, with a lack of microbial diversity, increased stress hormones, and consequently it becomes a driver of dementia. In fact, the National Institute on Aging published this year a longitudinal study that found that loneliness and isolation increased dementia risk by 31%. That is an equivalent increase to smoking. Chronic stress can really raise the risk of cognitive impairment. This is something that is driven by the enteric nervous system dysfunction and the dysbiosis and inflammation that result from that, affecting BDNF, which is responsible for the survival and the growth of these neural connections. A very large cohort study of over a million people that came out of Stockholm recorded chronic stress, depression, or both in, like I said, over a million people between 2012 and 2013. The outcome was a diagnosis. The outcome that was measured was a diagnosis of Alzheimer's disease, some other kind of dementia, or MCI, mild cognitive impairment recorded from 2014 through 2022. Turns out that there is an increased risk for developing memory impairment that is statistically significant. Alzheimer's, dementia, or mild cognitive impairment if you have any of these diagnoses of chronic stress, depression, or anxiety, Social determinants of health seem to set one up for a future of irritable bowel syndrome. Social determinants of health impact all areas of the DIGIN mnemonic in the context of irritable bowel, maternal stress level, mode of delivery, early nutrition, adverse childhood events, all can be considered antecedents to IBS. This clinical example, the gut-brain axis on the timeline shows us an individual who has developed a GBA or gut-brain axis dysfunction, and I'm gonna ask you, how would you tell this story back to your patient? Sometimes it can be really difficult to communicate these complicated concepts to our patients. So I'm going to do a little role play here and try to show you how I would do that, how I would communicate gut-brain axis dysfunction in simple terms, okay? So here we go. You ready? Thank you for coming in today. I understand that you're looking for some help with your anxiety and depression and your joint pain. I'm gonna attempt to show you how your current health concerns are connected to your childhood, your stress level, and your early life traumas and diet. Please feel free to correct me if I get any of your history wrong. You grew up in different apartments with a single mom who wasn't always employed, so you moved around a lot. You were really stressed by this instability as a child. She had many boyfriends that were not always kind to you, and you were often left by yourself. You had many ear infections as a baby that required antibiotics, and your mom, although she fed you as well as she could, often fed you packaged and processed food. At an early age, the microbes in your gut were really influenced by those antibiotics and poor diet quality, causing the decline of your healthy gut bacteria and altering your microbiome in a way that promoted hormone imbalance, food cravings, and poor digestion. In your teenage years, you experienced some very stressful events that impacted your nervous system in a negative way. Traumas stay in the body for a long time after we feel like we've dealt with

them intellectually. This impact to your nervous system also affected the nervous system in your gut. This nervous system is called the enteric nervous system. There's a line of communication like a telephone wire between your brain and your digestive system. After these events, you were on high alert. You didn't trust your dating partners, you didn't trust them to be kind to you. Your early childhood experiences, or what we call ACEs of adverse childhood events, also wired part of your nervous system called the vagus nerve to function in a way that promoted motility issues. You've told me that you played soccer up until the age of 16, which you loved, but you suffered a concussion after a collision with another player. Evidence tells us that concussions can change your gut microbiome in a negative way. You quit soccer after that, but then you started to gain weight and suffer from constipation. Likely a direct result of this enteric nervous system change from the trauma in your early life and the change in your microbiota from the concussion. In your 30s, after your children were born, you started jogging to lose weight, but you missed a curb and you broke your ankle. Since then, you're less inclined to exercise because your ankle hurts and it's stiff in the morning. You had a stressful marriage that ended in your 40s. You had to work more hours to make ends meet, and you developed insomnia. This contributes to more weight gain because you had no energy or time to prepare healthy food or to exercise. Soon after, you developed sleep apnea and gut disease. When I look at your timeline, I can see that your gut microbiome was altered in infancy, setting you up for some of your future health concerns, but really the chronic stress throughout your life, the long work hours, the failing marriage, the housing insecurity as a child, the trauma as a teenager caused further disruption in your gut via this enteric nervous system. When we eat when we're stressed or rushed, the fight or flight nervous system is dominant and the entire system is affected as a result. We're meant to eat in a calm, relaxed state called rest and digest to optimize all of the chemicals in the hormones of digestion. Your brain and your gut, they talk to each other every day all day. They rely on each other. The enteric nervous system has cells that make hormones that can influence your mood, your pain levels, think about your sore knees or your ankle, and your metabolism, causing weight gain. And this system talks to other areas of your body as well, causing you sometimes to make poor food choices, have food cravings, and gain weight. Some of the hormones in your gut can be affected by this history and can further your symptoms of anxiety, depression, and joint pain, it's all connected. So that is the gut-brain axis, the enteric nervous system on a timeline of an example patient that came into my office. I hope that was helpful in retelling the story, you can hear how I would communicate that to the patient. So, in summary, what are some of our key takeaways? We started this discussion talking about the gut-brain axis, you know, which is the highway, the big freeway, the enteric nervous system, and the vagus nerve. We looked at normal physiology of the gut-brain axis, and we acknowledged that it can function independently of the central nervous system. We reviewed functions

like propulsion, barrier integrity, immunity, and the bidirectionality of the signaling. We noted that the vagus nerve directly connects to the GI tract and to the central nervous system, but it does not come into contact with microbiota. We learned about the enteroendocrine system because of all of the important hormones secreted by these cells. And then we looked at the clinical findings associated with gut-brain axis dysfunction, and acknowledged that both the gut-brain axis and the enteric nervous system are mediators. We especially focused on the stressors to the gut-brain axis as mediators of enteric nervous system dysfunction. So this slide, Clinical Findings of GI dysfunction, is here to help you recognize the clinical indicators of enteric nervous system dysfunction specific to dysmotility. It is there for your reference. This slide is a continuation of that information. GI dysfunctions are mediated, as you know, by the enteric nervous system and the gut-brain axis, and this chart and the next one are just summary slides so you have a reference point. So finally, please remember that these modifiable lifestyle factors are all mediators to the enteric nervous system. Modifiable lifestyle factors are the low hanging fruit in these cases that you can help your patients modify. This is a very influential area of impact when you're working with patients on restoring their gut function. Being able to recognize the intestinal manifestations and the extraintestinal manifestations of these stressors is crucial in restoring proper gut function. When these lifestyle factors are out of balance, sleep, exercise, diet, mental health, relationships, all of the sections of the DIGIN mnemonic are impacted. Digestion, absorption, intestinal permeability, gut microbiota, the immune system and inflammation, and of course the nervous system. I hope you enjoyed the presentation, I'll see you soon.